

**SOMAIYA**

VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

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Roll Number: 09

Batch: 1

Experiment performed on (date): 04/09/2026

Title of the Experiment: Grating Constant

Aim: To determine the wavelength (λ) of He-Ne laser light.

Apparatus: He-Ne Laser apparatus, Grating Scale.

Observations:

Table 1:

Screen Distance $D = 10\text{cm}$

Grating Type - 15000

Sr No.	Order	Distance between Grating and Screen in cm(D)	Left (y_1) cm	Right (y_2) cm	Mean $y = \frac{y_1 + y_2}{2}$	$\theta = \tan^{-1} \frac{y}{D}$	$\sin \theta$	$\frac{\lambda (\cos \theta \sin \theta)}{n}$
1	1	10	5.164	5.164	5.164	27.31°	0.4588	776.9 nm
2	2	10	10.7	10.7	10.7	45.84°	0.7173	607.3 nm
3	3	10	15.6	15.6	15.6	57.33°	0.8417	475.0 nm

Table 2:

Screen Distance $D = 10\text{cm}$

Grating type - 10000

1	1	10	3.440	3.440	3.440	18.98°	0.3252	549.5 nm
2	2	10	6.892	6.892	6.892	34.57°	0.5674	476.4 nm
3	3	10	10.36	10.36	10.36	46.01°	0.7194	405.2 nm

Calculations:

$$\text{Avg}_1 = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3} = \frac{776.9 + 607.3 + 475.0}{3} = 619.7$$

$$\text{Avg}_2 = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3} = \frac{549.5 + 476.4 + 405.2}{3} = 477.0$$



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$$\text{Exp } \lambda = \frac{619.7 + \text{Avg}_1 + \text{Avg}_2}{2} = \frac{619.7 + 477}{2} = 550.36 \text{ nm}$$

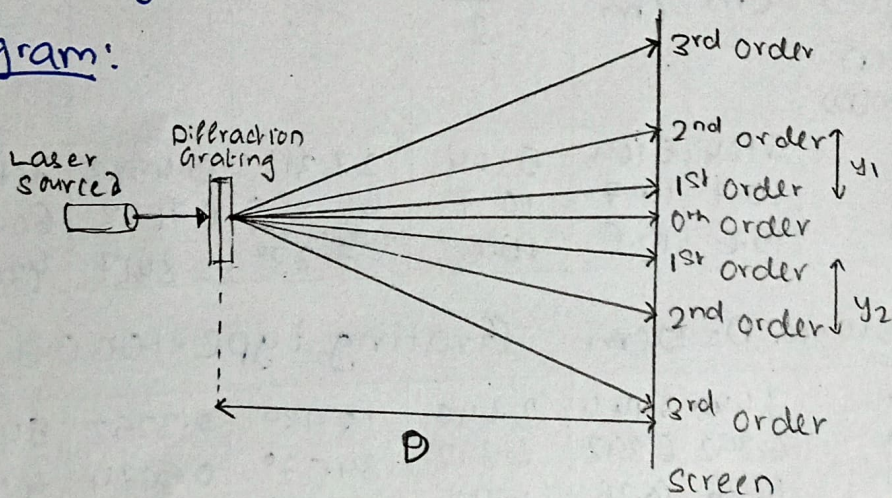
$$\text{Standard } \lambda = 600 \text{ nm}$$

$$\text{Percent Error} = \left| \frac{550.36 - 600}{600} \right| \times 100 = 8.27\%$$

Result/s and Conclusion/s:

1. Wavelength (λ) of a given light is 550.36 nm
2. Standard value of wavelength (λ) of a given light is 600 nm.
3. Percentage Error in the obtained result 8.27%.

Diagram:



Assignment:

1. Explain the principle of diffraction of light using a plane diffraction grating, and define the grating element ($a+b$).
- A plane transmission grating consists of a very large number of identical, equally spaced, parallel slits. When monochromatic light falls normally on it, each slit diffracts the light and the wavelets from all slits superpose. The resultant intensity is the product of single-slit diffraction and multiple slit interference, producing a few extremely sharp, bright principal maxima at angles satisfying the grating equation $(a+b)\sin\theta = n\lambda$ where $n = 0, 1, 2$ is the order. These maxima are narrow and intense, with negligible secondary maxima.

grating element $(a+b)$: It is the distance between corresponding points on two consecutive slits - the width a of one transparent slit plus the width b of one opaque adjacent ruled line. It is the repeat distance of the grating, and for n lines per unit length, $a+b = \frac{1}{n}$

2. He-Ne laser beam of wavelength $= 632.8 \text{ nm}$ is incident normally on a diffraction grating with a grating element $(a+b) = 1.693 \times 10^{-6}$. Calculate the angle of diffraction for the 2nd order diffraction maximum $n=2$.

$$\begin{aligned} \rightarrow \lambda &= 632.8 \text{ nm} && \text{(Wavelength)} \\ (a+b) &= 1.693 \times 10^{-6} && \text{(Grating element)} \\ n &= 2 && \text{(Order)} \end{aligned}$$

$$\lambda = \frac{(a+b) \sin \theta}{n}$$

$$632.8 \times 10^{-9} = \frac{1.693 \times 10^{-6} \times \sin \theta}{2}$$

$$\sin \theta = \frac{632.8 \times 10^{-9} \times 2}{1.693 \times 10^{-6}}$$

$$\sin \theta = 747.5 \times 10^{-3}$$

$$\begin{aligned} \theta &= \sin^{-1}(0.7475) \\ &= 48.37^\circ \end{aligned}$$

The angle of diffraction for the 2nd order maximum is 48.4° .